SUBAERO: EXHAUSTIVE END-TO-END MASTER TECHNICAL DOCUMENTATION

Physics-Informed Digital Twin for Real-Time Four-Stage Turbojet Health Monitoring

Problem Statement: PS-01 (IIT Indore & Hindustan Aeronautics Limited - HAL)
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Repository: https://github.com/prajansanjayk1/subaero.git | **Classification:** Master Technical Reference (12-15 Pages)

1. Executive Summary & System Scope

Modern gas turbine aerospace propulsion systems operate under severe thermal, mechanical, and aerodynamic gradients. Over extended operational cycles, critical components experience degradation such as compressor blade fouling, tip clearance opening, combustor liner thermal cracking, and turbine blade creep erosion. Because critical internal component states cannot be directly measured with physical flight sensors during operation, aerospace operators require Digital Twin technology capable of continuously estimating hidden engine states, predicting net thrust force and specific fuel consumption (TSFC), and forecasting remaining operational lifespan.

SubAero is a 100% White-Box Physics-Informed Digital Twin built specifically for four-stage single-spool turbojet engines. By decoupling fundamental gas dynamics thermodynamics from statistical corrections, SubAero delivers real-time health estimation and performance surrogate modeling with 0ms client-side latency and an average dataset accuracy of 98.89% across 30,000 flight cycles.

2. Problem Statement (PS-01) & Aerodynamic Flight Degradation Physics

Problem Statement 01 (PS-01), sponsored by IIT Indore and HAL, requires developing an interpretable, real-time Digital Twin for a four-stage turbojet engine operating under varying flight conditions. Telemetry streams contain 12 raw operational channels: Altitude, Mach number, Ambient Temperature (Tamb), Ambient Pressure (Pamb), Shaft Speed (RPM), Fuel Flow Rate, Compressor Inlet Pressure (P2), Compressor Inlet Temperature (T2), Combustor Inlet Pressure (P3), Combustor Inlet Temperature (T3), Turbine Exit Pressure (P4), and Turbine Exit Temperature (T4).

Real Flight Anomaly Scenario (Engine #38 Cruise Flight): Consider Engine #38 flying at cruise altitude of 4,452 meters, Mach 0.7257, and shaft speed 50,649 RPM. Telemetry records combustor exit temperature T3 at 714.98 K and turbine exit T4 at 560.5 K. Standard threshold alarms indicate safe temperatures. However, hidden aerodynamic blade fouling has dropped compressor isentropic efficiency by 10.4%, reducing net thrust to 15,277 Newtons while spiking fuel consumption. SubAero reconstructs this hidden degradation instantly.

3. Key Quantified Project Outcomes & Master Accuracy Matrix

SubAero achieves state-of-the-art accuracy across all six primary targets evaluated on a 30,000-row benchmark dataset (100 turbofan engines, 24,000 training rows / 6,000 held-out test rows):

Target Parameter / Output	Surrogate Model Engine	Test Dataset Accuracy	Test MAE / RMSE	R^2 Score
TSFC (Fuel Consumption)	Degree-2 Polynomial Ridge	**99.97%**	0.000253 g/(N*s) | 0.000339	0.9969
Thrust Force (N)	Degree-2 Polynomial Ridge	**99.27%**	439.98 N | 572.33 N	0.9989
Combustor Health	Degree-2 Polynomial Ridge	**98.98%**	0.010233 | 0.013416	0.7276
Overall Engine Health	Degree-2 Polynomial Ridge	**98.77%**	0.012305 | 0.016335	0.8808
Compressor Health	Degree-2 Polynomial Ridge	**98.30%**	0.016984 | 0.023027	0.8816
Turbine Health	Degree-2 Polynomial Ridge	**98.02%**	0.019784 | 0.025685	0.7392

4. Decoupled 3-Layer Hybrid System Architecture

SubAero uses a 3-layer hybrid architecture that decouples fundamental gas dynamics from statistical corrections:
1. **Layer 1: First-Principles Gas Dynamics Engine**: Computes Brayton cycle efficiency, pressure ratios, temperature ratios, and thermodynamic work directly from physical laws.
2. **Layer 2: 100% White-Box Polynomial Ridge Surrogate**: Evaluates 91 explicit linear, quadratic, and cross-sensor interaction terms. Every calculation is a transparent closed-form algebraic matrix dot product.
3. **Layer 3: Bidirectional Physics Safety Guardian**: Automated safety watchdog enforcing thermodynamic feasibility rules ($T3 > T2$, $T4 < T3$, $P3 > 1.05*P2$, $EGT \le 1273.15$ K). Enforces physical clamping if sensor telemetry is corrupted.

5. First-Principles Gas Dynamics Physics Implementation (Layer 1)

Layer 1 derives thermodynamic state variables directly from conservation of energy and momentum:

- **Ideal Compressor Exit Temperature (T2_is)**: $T2_{is} = T_{amb} * (P3 / P2)^{((\gamma-1)/\gamma)}$ for $\gamma = 1.4$
- **Compressor Isentropic Efficiency (eta_c)**: $\eta_c = (T2_{is} - T_{amb}) / (T2 - T_{amb})$
- **Combustor Temperature Ratio (TR)**: $TR = T3 / T2$
- **Turbine Expansion Work Coefficient (W)**: $W = (T3 - T4) / T3$
- **Thermal Stress Index (sigma)**: $\sigma = (T3 / T2) * (P3 / P2)$

6. 100% White-Box Polynomial Matrix Surrogate Engine (Layer 2)

Layer 2 replaces black-box neural networks with explicit L2-regularized Polynomial Matrix dot products. For 12 normalized telemetry inputs $z = [z1, ..., z12]^T$, the 91 polynomial terms comprise:

- 1 Intercept Bias Term ($w0$)
- 12 Linear Feature Terms (z_i)
- 12 Quadratic Feature Terms (z_i^2)
- 66 Cross-Sensor Interaction Terms ($z_i * z_j$ for $1 <= i < j <= 12$)

Equation: $\hat{y} = w0 + sum(w_i * z_i) + sum(w_{ij} * z_i * z_j)$

7. Layer 3 Bidirectional Physics Safety Guardian Rules

Layer 3 acts as an automated safety watchdog intercepting predictions that violate thermodynamic physical laws:

Physical Rule	Mathematical Boundary	Enforcement Action
Compressor Energy Addition	T3 > T2	Flag Compressor_Work_Inversion alert and clamp T3
Combustor Thermal Expansion	T4 < T3	Flag Combustor_Heat_Inversion alert and clamp T4
Compressor Pressure Ratio	P3 > P2 * 1.05	Flag Compression_Loss_Surge alert
Turbine Blade Melt Limit	EGT <= 1273.15 K	Flag EGT_CRITICAL_OVERTEMP emergency alert
Physical Health Scale	H in [0.10, 0.9999]	Clamp H = min(0.9999, max(0.10, y_hat))

8. 11-Node Hierarchical Subsystem Health Tree Decomposition

Overall engine health is decomposed into an 11-node tree:

• Overall Engine Health (100% aggregate)

■■■ Mechanical Health (40% weight): Compressor Health (50%) & Turbine Health (50%)

■■■ Thermal Health (25% weight): EGT Margin Health & Combustor Thermal State

■■■ Pressure Health (20% weight): Compressor PR Health & Turbine PR Health

■■■ Combustion Health (10% weight): Combustor Health

■■■ Efficiency Health (5% weight): Isentropic Efficiency & Turbine Work Coefficient

9. Remaining Useful Life (RUL) Degradation Trajectory Engine

RUL prediction uses health degradation tracking rather than cycle counting:

• Exponential Moving Average: $EMA_{10}(t) = 0.20 * H_{overall}(t) + 0.80 * EMA_{10}(t-1)$

• Degradation Slope: $dH/dt = (EMA_{10}(t) - EMA_{10}(t-10)) / 10$

• Projected RUL: Estimated $RUL = (EMA_{10}(t) - 0.70) / (|dH/dt| + 1e\text{-}9)$

Alert Status: Green Normal (>=150 cycles), Yellow Monitor (80-150 cycles), Orange Warning (30-80 cycles), Red Critical (<30 cycles).

10. Leave-One-Engine-Out (LOEO) Cross-Validation Matrix

Engine Fold Group	Overall Health MAE	Thrust MAE (N)	TSFC MAE	Fold R^2 Score
Engines 1 - 20 (Test Set)	0.01231	439.98 N	0.000253	0.8808
Engines 21 - 40	0.01245	445.12 N	0.000256	0.8791
Engines 41 - 60	0.01218	432.45 N	0.000249	0.8835
Engines 61 - 80	0.01238	441.04 N	0.000254	0.8802
Engines 81 - 100	0.01222	435.30 N	0.000251	0.8824
LOEO Mean Total	**0.01231**	**438.78 N**	**0.000253**	**0.8812**

11. Single Sample Evaluation vs Full Dataset Benchmark

Single Sample Evaluation (Engine #38, Cycle 177):
• Overall Health: Actual = 91.25%, Predicted = 92.14% (99.12% Accuracy)
• Compressor Health: Actual = 89.96%, Predicted = 89.56% (99.60% Accuracy)
• Combustor Health: Actual = 88.00%, Predicted = 96.06% (91.94% Accuracy)
• Turbine Health: Actual = 96.79%, Predicted = 91.65% (95.41% Accuracy)
• Thrust Force: Actual = 15,277.93 N, Predicted = 16,648.72 N (97.72% Accuracy)
• TSFC: Actual = 0.014223 g/(N*s), Predicted = 0.013876 g/(N*s) (99.31% Accuracy)
• **Single Sample Average Accuracy = 97.18%**

12. 0ms Latency Client-Side Browser Matrix Execution Architecture

SubAero serializes model weights, means, scales, intercepts, and power matrices (15.2 KB) into src/assets/whitebox_models_12sensors.json. The TypeScript engine calculates predictions natively inside the React frontend browser engine with **0ms server latency**.

```
predict_single.py - Standalone CLI predictor for 12 telemetry sensors.
BatchExcelAccuracyCalculator.tsx - Drag-and-drop CSV validation tool.
SubAero_Null_Pointers_Final_Report.tex - Official 15-page LaTeX technical report for PS-01.
```

13. Literature Survey Comparison & State-of-the-Art Benchmarks

Evaluation Parameter	Literature Benchmark (Elsevier 2025 & NASA)	SubAero (Team Null Pointers)
Model Interpretability	Black-box neural networks (CNNs, LSTMs)	**100% White-Box Polynomial Matrix Equations**
Inference Latency	50ms - 200ms server HTTP request delay	**0ms client-side browser matrix execution**
Physics Integration	Offline residual loss penalties	**Layer 3 Real-Time Physics Safety Guardian**
Generalization	Single engine split validation	**LOEO cross-validation across 100 engines (R^2=0.8812)**

14. Team Null Pointers Roles & Enterprise Deployment Roadmap

Team Composition:
• **Prajan Sanjay K**: Lead Digital Twin Architect (3D twin development in Blender & Unity integration).
• **Sridharshini S**: Physics Modeling Engineer (Brayton-cycle physics engine & thermodynamic equations).
• **Kishore Kumar P**: Machine Learning Engineer (100% White-Box Polynomial Ridge model pipeline & LOEO validation).
• **Nithish Bharathwaj N**: Full-Stack Software Engineer (React Mission Control Dashboard & edge API integration).

Enterprise Deployment Roadmap:
Phase 1: Benchmarking & LOEO Audits (Done) -> Phase 2: On-Prem Edge Gateway & MIL-STD-1553 Adapter -> Phase 3: FAA / EASA Digital Twin Certification.

15. Conclusion & Competition Deliverables Checklist

SubAero fulfills all requirements of Problem Statement 01:
1. **Health Estimation Accuracy (30%)**: 98.02% - 98.98% Accuracy across all health targets.
2. **Surrogate Model Performance (20%)**: 99.27% Thrust Accuracy and 99.97% TSFC Accuracy.
3. **Physics Consistency (15%)**: 100% adherence to gas dynamics equations and Layer 3 Physics Guardian.
4. **Generalization Capability (15%)**: LOEO cross-validation confirms 0.8812 mean R^2 across unseen engines.
5. **Computational Efficiency (10%)**: 0ms server latency, 15.2 KB client-side asset.
6. **Dashboard & Interpretability (10%)**: Feature attribution suite and interactive UI.